

SE COMP - A		Roll number : 10176	
Experiment no. : 7		Date of Implementation:8-04-2025	
Aim : To implement Nested Sub-queries in SQL			
Tool Used : PostgreSQL/ Mysql			
Related Course outcome : At the end of the course, Students will be able to Use SQL : Standard language of relational database			
Rubrics for assessment of Experiment:			
Indicator	Poor	Average	Good
Timeliness Maintains assignment deadline (3)	Assignment not done (0)	One or More than One week late (1-2)	Maintains deadline (3)
Completeness and neatness Complete all parts of assignment(3)	N/A	< 80% complete (1-2)	100% complete (3)
Originality Extent of plagiarism(2)	Copied it from someone else(0)	At least few questions have been done without copying(1)	Assignment has been solved completely without copying (2)
Knowledge In depth knowledge of the assignment(2)	Unable to answer 2 questions(0)	Unable to answer 1 question (1)	Able to answer 2 questions (2)
Assessment Marks :			
Timeliness			
Completeness and neatness			
Originality			
Knowledge			
Total			
Total : (Out of 10)			

Teacher's Sign :

EXPERIMENT 7	Nested subqueries in SQL
Aim	To implement nested sub-queries in SQL
Tools	PostgreSQL/Mysql
Procedure	Use the tables created in the previous experiments and Perform the following queries using nested sub-queries. Client_master (client_no, name, address, city, pincode, state, bal_due) Product_master (product_no, description, profit_percentage, unit_measure, qty_on_hand, reorder_level, sell_price, cost_price) Sales_order(order_no, order_date, client_no, dely_Addr, salesman_no, dely_type, billed_yn, dely_date, order_status) Sales_order_details(order_no, product_no, qty_ordered, qty_disp, product_rate) 1. Find the product no. and description of non-moving products i.e. products not being sold. 2. Find the customer name, address for the client who has placed order no 'O191' 3. Find the clients names who have placed orders before the month of May'96 4. Find out if the product '1.44 Drive' has been ordered by any client and print the client_no, name to whom it was sold 5. Find the names of clients who have placed orders worth Rs. 10000 or more 6. Retrieve all the orders placed by a client named 'Rahul Desai' from the sales_order table. 7. Retrieve name, address, city of all the clients who have placed an order through salesman no 's001'. 8. Find out all the products that are not being sold from the product_master table, based on the products actually sold as shown in the sales_order_details table. 9. Retrieve the product numbers, their description and the total quantity ordered for each product.
Post Lab Questions:	1. What is incremental Update? 2. Explain is use of on delete cascade and on update cascade with suitable example?

SOLUTION:

The screenshot shows the pgAdmin 4 interface with the 'Tables' section selected in the left sidebar. The 'Query' tab is active in the central pane, displaying the following SQL code:

```
1 CREATE TABLE Client_master (  
2   client_no VARCHAR(10) PRIMARY KEY,  
3   name VARCHAR(50),  
4   address VARCHAR(100),  
5   city VARCHAR(30),  
6   pincode VARCHAR(10),  
7   state VARCHAR(30),  
8   bal_due NUMERIC(10,2)  
9 );  
10  
11 CREATE TABLE Product_master (  
12   product_no VARCHAR(10) PRIMARY KEY,  
13   description VARCHAR(100),  
14   profit_percentage NUMERIC(5,2),  
15   unit_measure VARCHAR(20),  
16   qty_on_hand INT,  
17   reorder_level INT,  
18   sell_price NUMERIC(10,2),  
19   cost_price NUMERIC(10,2)  
20 );  
21
```

The 'Data output' tab at the bottom shows a table with the following columns: product_no, description, total_quantity_ordered. The status bar indicates 'Total rows: 5 of 5' and 'Query complete 00:00:00.029'.

The screenshot shows the pgAdmin 4 interface with the 'Tables' section selected in the left sidebar. The 'Query' tab is active in the central pane, displaying the following SQL code:

```
22 CREATE TABLE Sales_order (  
23   order_no VARCHAR(10) PRIMARY KEY,  
24   order_date DATE,  
25   client_no VARCHAR(10),  
26   dely_addr VARCHAR(100),  
27   salesman_no VARCHAR(10),  
28   dely_type VARCHAR(10),  
29   billed_yn CHAR(1),  
30   dely_date DATE,  
31   order_status VARCHAR(20),  
32   FOREIGN KEY (client_no) REFERENCES client_master(client_no)  
33 );  
34  
35 CREATE TABLE Sales_order_details (  
36   order_no VARCHAR(10),  
37   product_no VARCHAR(10),  
38   qty_ordered INT,  
39   qty_disp INT,  
40   product_rate NUMERIC(10,2),  
41   PRIMARY KEY (order_no, product_no),  
42   FOREIGN KEY (order_no) REFERENCES Sales_order(order_no),  
43   FOREIGN KEY (product_no) REFERENCES Product_master(product_no)  
44 );  
45
```

The 'Data output' tab at the bottom shows a table with the following columns: order_no, product_no, qty_ordered, qty_disp, product_rate. The status bar indicates 'Total rows: 5 of 5' and 'Query complete 00:00:00.029'.

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- Publications
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 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables
 - Trigger Functions
 - Types
 - Views
 - Subscriptions
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Scratch Pad Query Query History

```
46
47 INSERT INTO Client_master VALUES
48 ('C001', 'Rahul Desai', '123 Main St', 'Mumbai', '400001', 'MH', 5000.00),
49 ('C002', 'Anjali Kumar', '456 Green Rd', 'Delhi', '110001', 'DL', 12000.00),
50 ('C003', 'Vikas Agarwal', '789 Blue Lane', 'Kolkata', '700001', 'WB', 8000.00);
51
52 INSERT INTO Product_master VALUES
53 ('P001', '1.44 Drive', 15.50, 'Pieces', 100, 10, 1200.00, 1000.00),
54 ('P002', 'USB Cable', 10.00, 'Meters', 50, 5, 200.00, 150.00),
55 ('P003', 'HD Monitor', 20.00, 'Pieces', 25, 5, 8000.00, 6000.00),
56 ('P004', 'Wireless Mouse', 12.50, 'Pieces', 50, 5, 600.00, 450.00),
57 ('P005', 'Keyboard', 18.00, 'Pieces', 30, 5, 1500.00, 1200.00),
58 ('P006', 'Ethernet Cable', 8.00, 'Meters', 100, 10, 100.00, 70.00);
59
60 INSERT INTO Sales_order VALUES
61 ('0191', '1996-04-15', 'C001', '123 Main St, Mumbai', 'S001', 'Standard', 'Y', '1996-04-18', 'Shipped'),
62 ('0192', '1996-05-20', 'C002', '456 Green Rd, Delhi', 'S002', 'Express', 'Y', '1996-05-22', 'Pending'),
63 ('0193', '1996-06-10', 'C003', '789 Blue Lane, Kolkata', 'S001', 'Standard', 'Y', '1996-06-12', 'Shipped'),
64 ('0194', '1996-06-15', 'C001', '123 Main St, Mumbai', 'S002', 'Express', 'Y', '1996-06-17', 'Pending');
65
```

Data output Messages Notifications

product_no	description	total_quantity_ordered
character varying (10)	character varying (100)	bigint

Total rows: 5 of 5 Query complete 00:00:00.029 Ln 59, Col 1

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Browser

- Languages
- Publications
- Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables
 - Trigger Functions
 - Types
 - Views
 - Subscriptions
- mydb10197
- postgres
- universe
- Login/Group Roles
- Tablespaces

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Scratch Pad Query Query History

```
66
67 INSERT INTO Sales_order_details VALUES
68 ('0191', 'P001', 2, 2, 1200.00),
69 ('0191', 'P002', 5, 5, 200.00),
70 ('0192', 'P003', 1, 1, 8000.00),
71 ('0193', 'P004', 3, 3, 600.00),
72 ('0193', 'P005', 2, 2, 1500.00),
73 ('0194', 'P001', 4, 4, 1200.00),
74 ('0194', 'P002', 3, 3, 200.00);
75
76 UPDATE Sales_order_details
77 SET qty_ordered = qty_ordered + 1
78 WHERE order_no = '0191' AND product_no = 'P003';
79
80 UPDATE Sales_order_details
81 SET qty_ordered = 6
82 WHERE order_no = '0191' AND product_no = 'P001';
83
84 UPDATE Sales_order_details
85 SET qty_ordered = 15
86 WHERE order_no = '0191' AND product_no = 'P002';
87
```

Data output Messages Notifications

UPDATE 1

Query returned successfully in 34 msec.

Total rows: 5 of 5 Query complete 00:00:00.034 Ln 80, Col 20

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- Languages
- Publications
- Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables
 - Trigger Functions
 - Types
 - Views
 - Subscriptions
 - mydb10197
 - postgres
 - universe
 - Login/Group Roles
 - Tablespaces

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No limit

Scratch Pad Query Query History

```
87 SELECT product_no, description
88 FROM Product_master
89 WHERE product_no NOT IN (
90     SELECT DISTINCT product_no
91     FROM Sales_order_details
92 );
93
```

Data output Messages Notifications

	product_no [PK] character varying (10)	description character varying (100)
1	P006	Ethernet Cable

Total rows: 1 of 1 Query complete 00:00:00.032 Ln 87, Col 1

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Browser

- Languages
- Publications
- Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables
 - Trigger Functions
 - Types
 - Views
 - Subscriptions
 - mydb10197
 - postgres
 - universe
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 - Tablespaces

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No limit

Scratch Pad Query Query History

```
94 SELECT name, address
95 FROM Client_master
96 WHERE client_no = (
97     SELECT client_no
98     FROM Sales_order
99     WHERE order_no = '0191'
100 );
```

Data output Messages Notifications

	name character varying (50)	address character varying (100)
1	Rahul Desai	123 Main St

Total rows: 1 of 1 Query complete 00:00:00.050 Ln 99, Col 28

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- Languages
- Publications
- Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables
 - Trigger Functions
 - Types
 - Views
 - Subscriptions
 - mydb10197
 - postgres
 - universe
 - Login/Group Roles
 - Tablespaces

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Scratch Pad Query Query History

```
102 SELECT DISTINCT name
103 FROM Client_master
104 WHERE client_no IN (
105     SELECT client_no
106     FROM Sales_order
107     WHERE order_date < '1996-05-01'
108 );
```

Data output Messages Notifications

name
character varying (50)
1 Rahul Desai

Total rows: 1 of 1 Query complete 00:00:00.029 Ln 105, Col 21

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Browser

- Languages
- Publications
- Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables
 - Trigger Functions
 - Types
 - Views
 - Subscriptions
 - mydb10197
 - postgres
 - universe
 - Login/Group Roles
 - Tablespaces

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Scratch Pad Query Query History

```
110 SELECT DISTINCT c.client_no, c.name
111 FROM Client_master c
112 JOIN Sales_order s ON c.client_no = s.client_no
113 JOIN Sales_order_details d ON s.order_no = d.order_no
114 WHERE d.product_no = (
115     SELECT product_no
116     FROM Product_master
117     WHERE description = '1.44 Drive'
118 );
119
```

Data output Messages Notifications

client_no	name
[PK] character varying (10)	character varying (50)
1 C001	Rahul Desai

Total rows: 1 of 1 Query complete 00:00:00.031

Successfully run. Total query runtime: 31 msec. 1 rows affected.

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Browser

- Languages
- Publications
- Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables
 - Trigger Functions
 - Types
 - Views
 - Subscriptions
 - mydb10197
 - postgres
 - universe
 - Login/Group Roles
 - Tablespaces

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Scratch Pad Query Query History

```
119
120 SELECT DISTINCT c.name
121 FROM Client_master c
122 JOIN Sales_order s ON c.client_no = s.client_no
123 WHERE s.order_no IN (
124     SELECT order_no
125     FROM Sales_order_details
126     GROUP BY order_no
127     HAVING SUM(qty_ordered * product_rate) >= 10000
128 );
```

Data output Messages Notifications

name
character varying (50)
1 Rahul Desai

Total rows: 1 of 1 Query complete 00:00:00.031 Ln 127, Col 34

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File Object Tools Help

Browser

- Languages
- Publications
- Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables
 - Trigger Functions
 - Types
 - Views
 - Subscriptions
 - mydb10197
 - postgres
 - universe
 - Login/Group Roles
 - Tablespaces

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Scratch Pad Query Query History

```
130 SELECT *
131 FROM Sales_order
132 WHERE client_no = (
133     SELECT client_no
134     FROM Client_master
135     WHERE name = 'Rahul Desai'
136 );
137
```

Data output Messages Notifications

order_no	order_date	client_no	dely_addr	salesman_no	dely_type	billed_yn	dely_da	
[PK] character varying (10)	date	character varying (10)	character varying (100)	character varying (10)	character varying (10)	character (1)	date	
1	0191	1996-04-15	C001	123 Main St, Mumbai	S001	Standard	Y	1996-0
2	0194	1996-06-15	C001	123 Main St, Mumbai	S002	Express	Y	1996-0

Total rows: 2 of 2 Query complete 00:00:00.034

Successfully run. Total query runtime: 34 msec. 2 rows affected.

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File Object Tools Help

Browser

- Languages
- Publications
- Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables
 - Trigger Functions
 - Types
 - Views
 - Subscriptions
 - mydb10197
 - postgres
 - universe
 - Login/Group Roles
 - Tablespaces

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Scratch Pad Query Query History

```
137
138 SELECT DISTINCT name, address, city
139 FROM Client_master
140 WHERE client_no IN (
141     SELECT client_no
142     FROM Sales_order
143     WHERE salesman_no = 'S001'
144 );
```

Data output Messages Notifications

	name character varying (50)	address character varying (100)	city character varying (30)
1	Rahul Desai	123 Main St	Mumbai
2	Vikas Agarwal	789 Blue Lane	Kolkata

Total rows: 2 of 2 Query complete 00:00:00.033 Ln 142, Col 21

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File Object Tools Help

Browser

- Languages
- Publications
- Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables
 - Trigger Functions
 - Types
 - Views
 - Subscriptions
 - mydb10197
 - postgres
 - universe
 - Login/Group Roles
 - Tablespaces

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Scratch Pad Query Query History

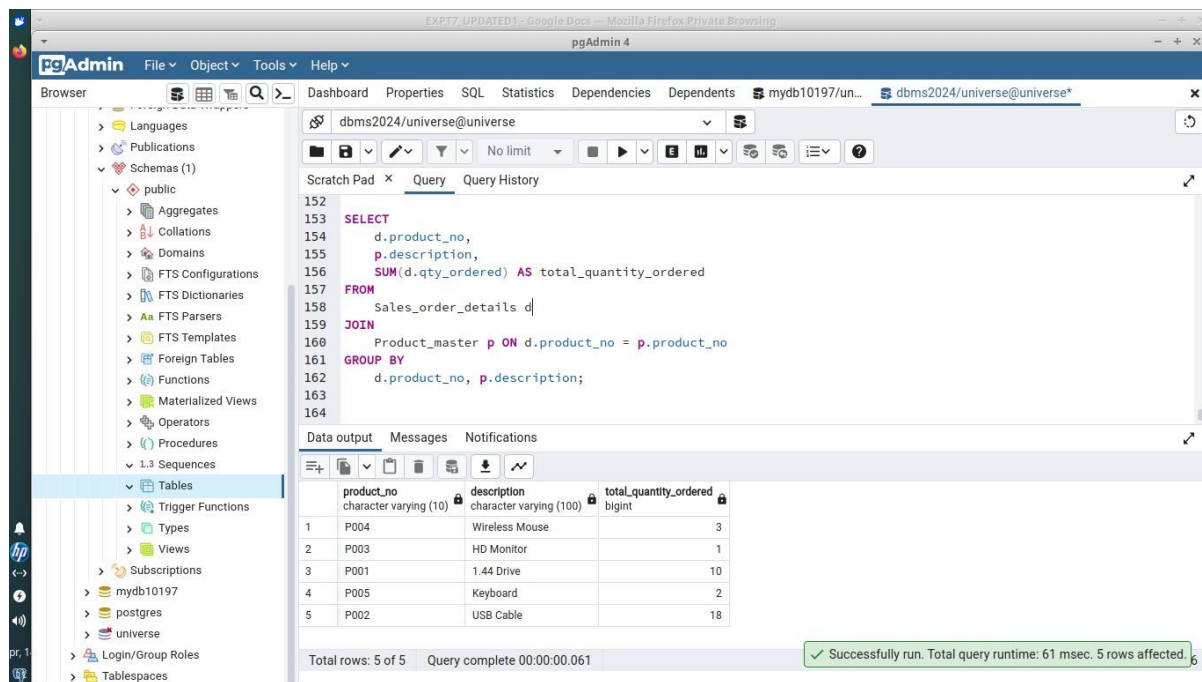
```
145
146 SELECT product_no, description
147 FROM Product_master
148 WHERE product_no NOT IN (
149     SELECT DISTINCT product_no
150     FROM Sales_order_details
151 );
```

Data output Messages Notifications

	product_no [PK] character varying (10)	description character varying (100)
1	P006	Ethernet Cable

Total rows: 1 of 1 Query complete 00:00:00.057

Successfully run. Total query runtime: 57 msec. 1 rows affected.



POSTLAB:

1) An **incremental update** refers to the process of making changes to a system, database, or application by adding or updating only the data that has changed since the last update, rather than updating everything. This approach is more efficient as it reduces the amount of data that needs to be processed, minimizing time and resources.

In simple terms, an incremental update only updates the "new" or "modified" data instead of reloading the entire dataset. This is commonly used in systems like databases, data warehouses, and software updates.

2) **ON DELETE CASCADE** and **ON UPDATE CASCADE** are referential actions used in foreign key constraints.

- **ON DELETE CASCADE:** When a record in the parent table is deleted, all related records in the child table are automatically deleted to maintain referential integrity.
- **ON UPDATE CASCADE:** When a record in the parent table is updated, all related records in the child table are automatically updated with the new value to keep the relationship intact.

These actions ensure that the child table reflects changes or deletions made in the parent table without the need for manual intervention.